DEPARTMENT OF
CIVIL AND
ENVIRONMENTAL
ENGINEERING

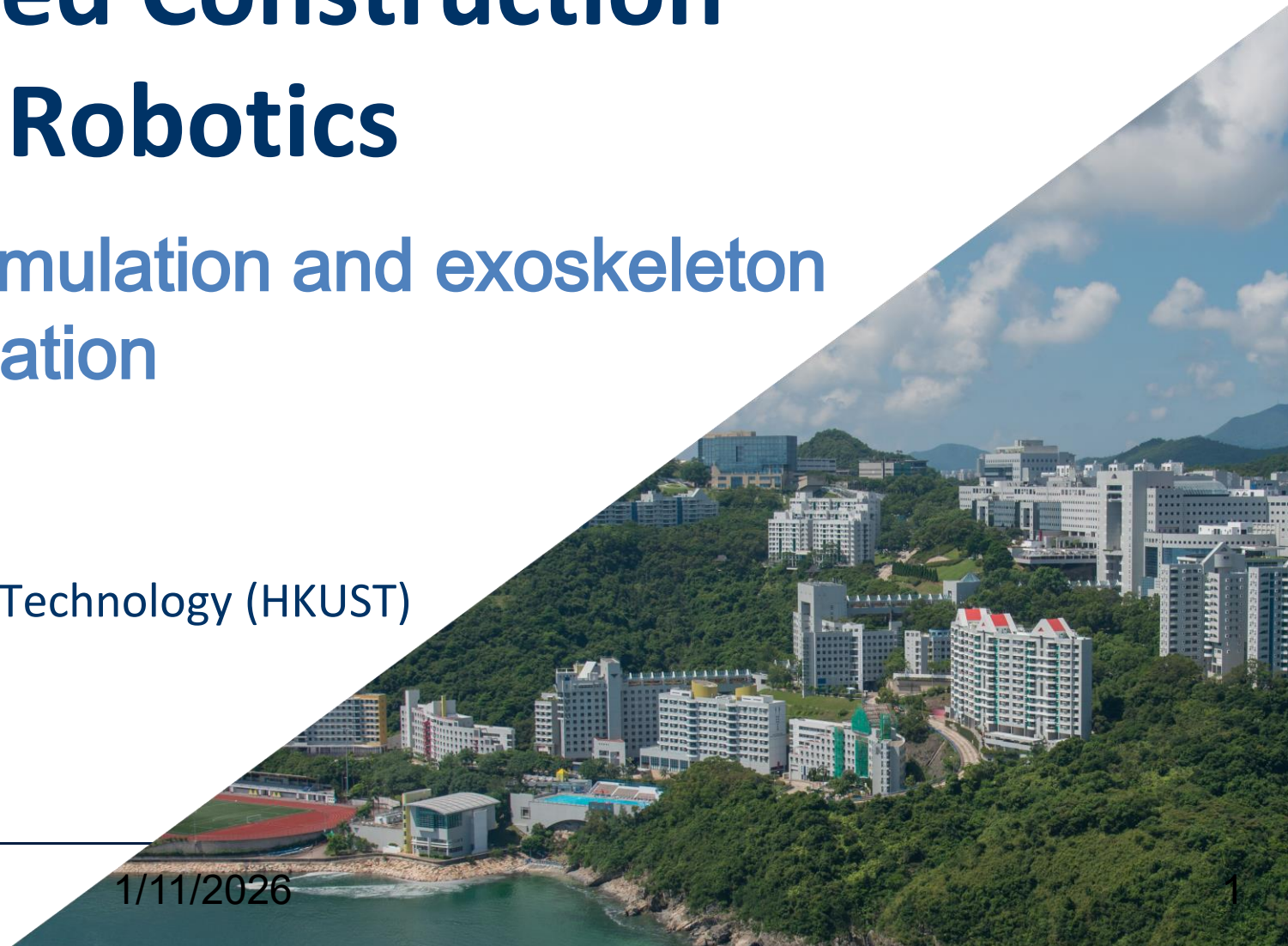
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THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

CIVL 4210 - Advanced Construction with AI and Robotics

Tutorial: Biomechanical simulation and exoskeleton simulation

Prof. YU, Yantao
XIE Minshan
The Hong Kong University of Science and Technology (HKUST)

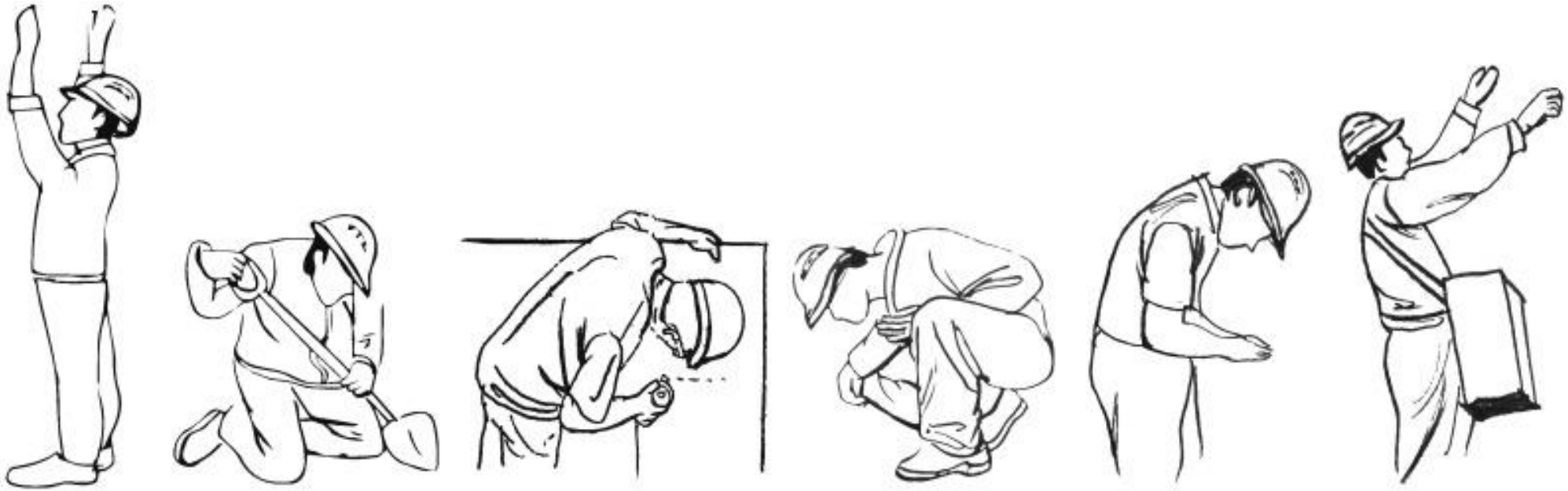


Introduction

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Construction work involves high physical demand and awkward postures



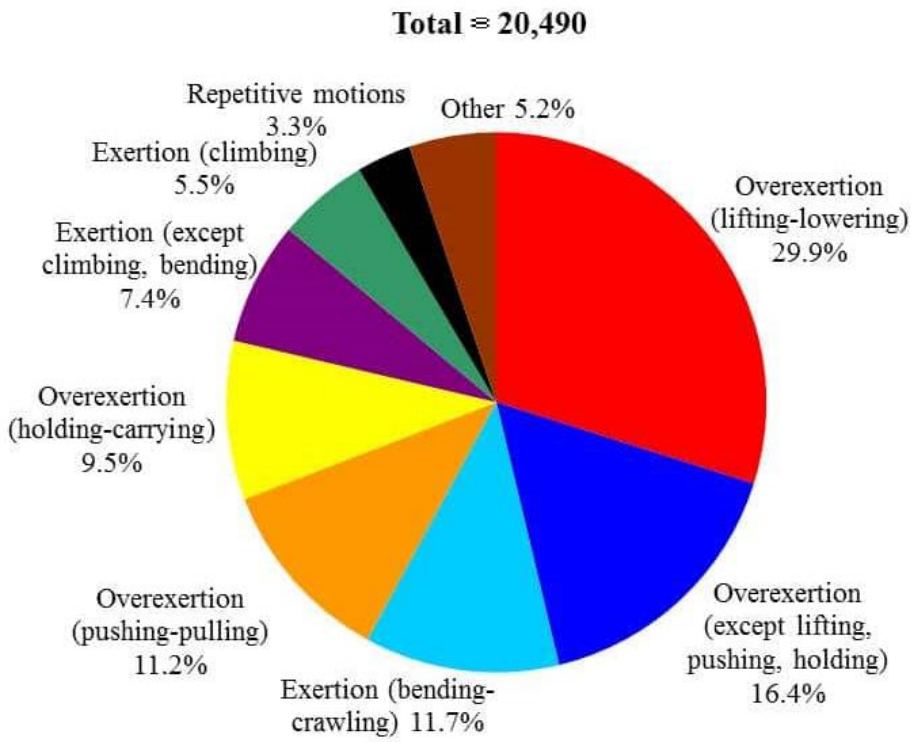
(a) Working Overhead (b) Kneeling (c) Back Bending Forward (d) Squatting (e) Neck Bending (f) Reaching

Introduction

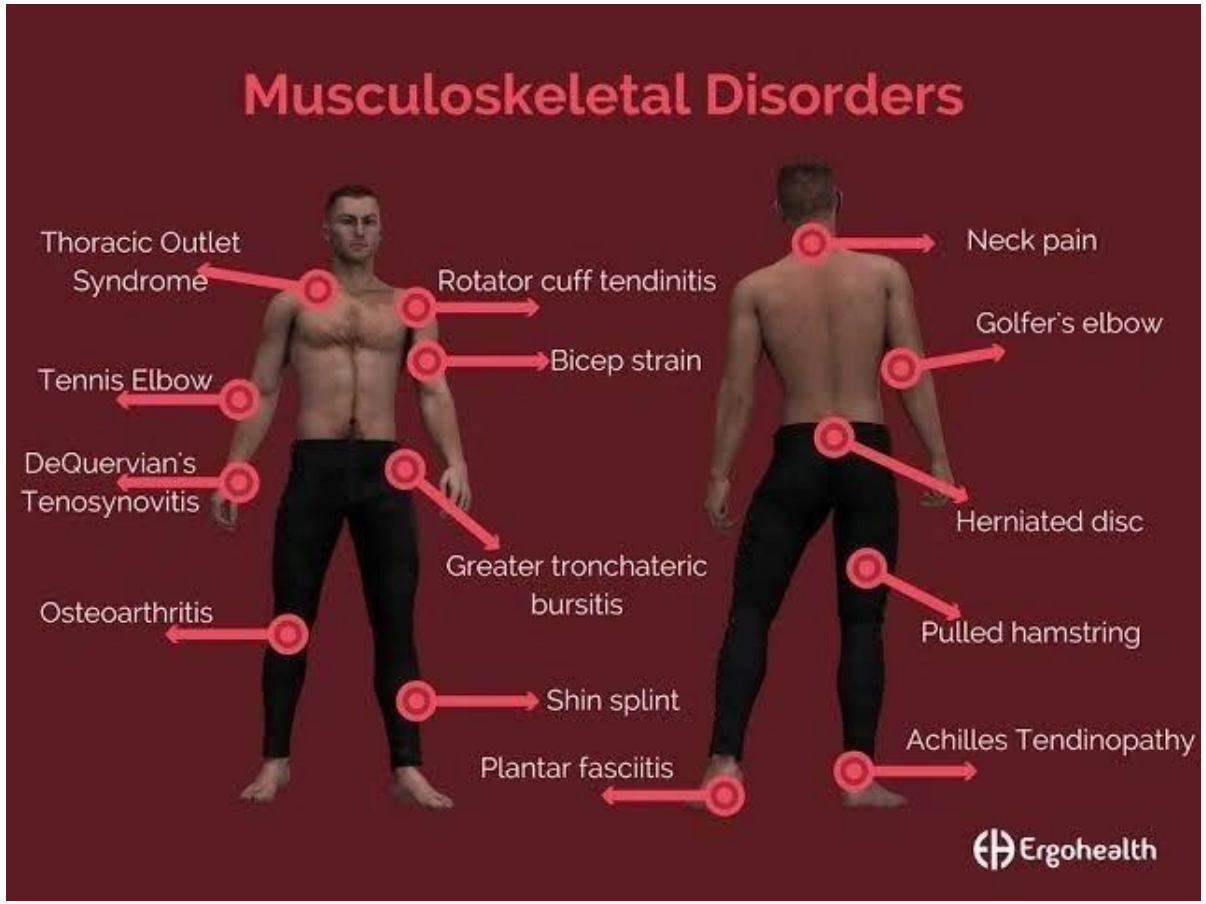
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Work-related musculoskeletal disorders (WMSDs) are physical conditions that cause pain in muscle and joints



"Other" includes overexertion and bodily reaction, unspecified; and multiple types of overexertion and bodily reactions.



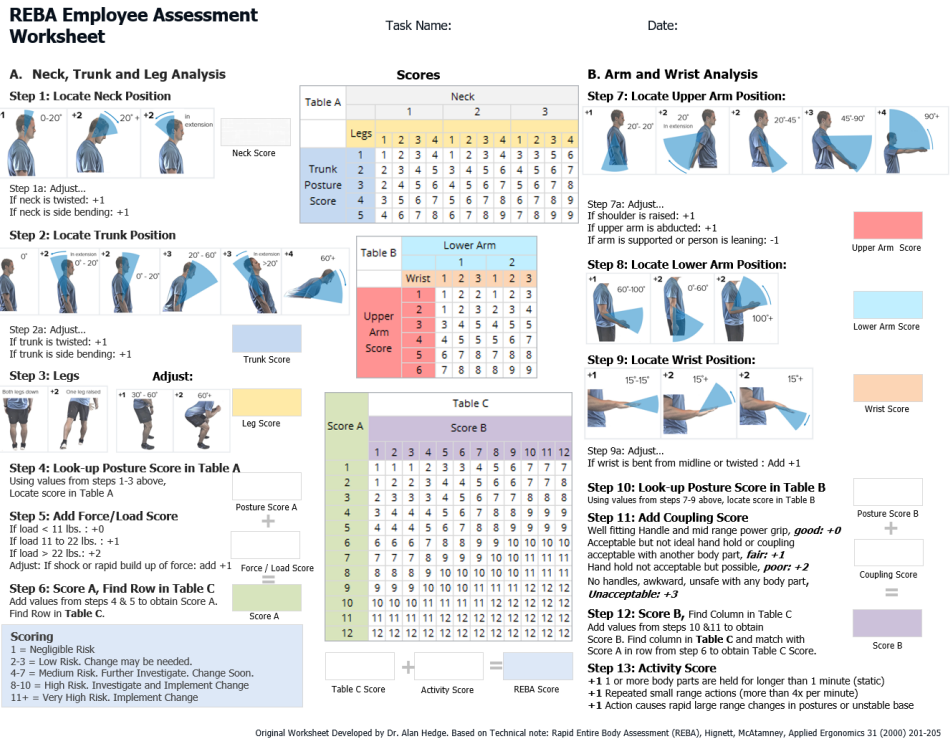
Major causes of WMSDs are internal physical strains, but how do we monitor them?

Introduction

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Survey-based approach



REBA Employee Assessment Worksheet

Task Name: Date:

A. Neck, Trunk and Leg Analysis

Step 1: Locate Neck Position

Step 1a: Adjust...
If neck is twisted: +1
If neck is side bending: +1

Step 2: Locate Trunk Position

Step 2a: Adjust...
If trunk is twisted: +1
If trunk is side bending: +1

Step 3: Legs

Adjust:

Step 4: Look-up Posture Score in Table A
Using values from steps 1-3 above, Locate score in Table A

Step 5: Add Force/Load Score
If load < 11 lbs. : +0
If load 11 to 22 lbs. : +1
If load > 22 lbs.: +2
Adjust: If shock or rapid build up of force: add +1

Step 6: Score A, Find Row in Table C
Add values from steps 4 & 5 to obtain Score A. Find Row in **Table C**.

Scoring
1 = Negligible Risk
2-3 = Low Risk. Change may be needed.
4-7 = Medium Risk. Further Investigate. Change Soon.
8-10 = High Risk. Investigate and Implement Change
11+ = Very High Risk. Implement Change

Neck Score | Trunk Score | Leg Score | Posture Score A | Force / Load Score | Score A

Scores

Table A

Trunk Posture Score	Neck 1				Neck 2				Neck 3			
	Legs 1	2	3	4	1	2	3	4	1	2	3	4
1	1	2	3	4	1	2	3	4	3	3	5	6
2	2	3	4	5	3	4	5	6	4	5	6	7
3	2	4	5	6	4	5	6	7	5	6	7	8
4	3	5	6	7	5	6	7	8	6	7	8	9
5	4	6	7	8	6	7	8	9	7	8	9	9

Table B

Upper Arm Score	Lower Arm 1 Wrist 1	2	3	Lower Arm 2 Wrist 1	2	3
1	1	2	2	1	2	3
2	1	2	3	2	3	4
3	3	4	5	4	5	5
4	4	5	5	5	6	7
5	6	7	8	7	8	8
6	7	8	8	8	9	9

Table C

Score A	Score B 1	2	3	4	5	6	7	8	9	10	11	12
1	1	1	1	2	3	3	4	5	6	7	7	7
2	1	2	2	3	4	4	5	6	6	7	7	8
3	2	3	3	3	4	5	6	7	7	8	8	8
4	3	4	4	4	5	6	7	8	8	9	9	9
5	4	4	4	5	6	7	8	8	9	9	9	9
6	6	6	6	7	8	8	9	9	10	10	10	10
7	7	7	7	8	9	9	9	10	10	11	11	11
8	8	8	8	9	10	10	10	10	10	11	11	11
9	9	9	9	10	10	10	11	11	11	12	12	12
10	10	10	10	11	11	11	11	12	12	12	12	12
11	11	11	11	11	12	12	12	12	12	12	12	12
12	12	12	12	12	12	12	12	12	12	12	12	12

Table C Score | Activity Score | REBA Score

B. Arm and Wrist Analysis

Step 7: Locate Upper Arm Position:

Step 7a: Adjust...
If shoulder is raised: +1
If upper arm is abducted: +1
If arm is supported or person is leaning: -1

Step 8: Locate Lower Arm Position:

Step 9: Locate Wrist Position:

Step 9a: Adjust...
If wrist is bent from midline or twisted : Add +1

Step 10: Look-up Posture Score in Table B
Using values from steps 7-9 above, locate score in Table B

Step 11: Add Coupling Score
Well fitting Handle and mid range power grip, ***good: +0***
Acceptable but not ideal hand hold or coupling acceptable with another body part, ***fair: +1***
Hand hold not acceptable but possible, ***poor: +2***
No handles, awkward, unsafe with any body part, ***Unacceptable: +3***

Step 12: Score B, Find Column in Table C
Add values from steps 10 &11 to obtain Score B. Find column in **Table C** and match with Score A in row from step 6 to obtain Table C Score.

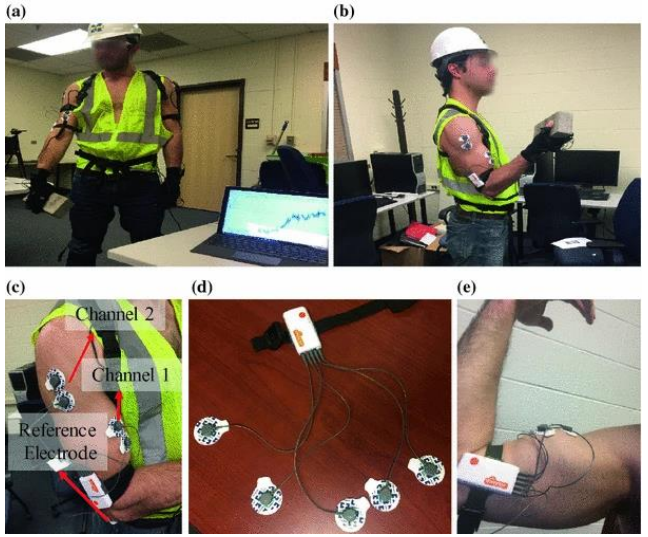
Step 13: Activity Score
+1 1 or more body parts are held for longer than 1 minute (static)
+1 Repeated small range actions (more than 4x per minute)
+1 Action causes rapid large range changes in postures or unstable base

Upper Arm Score | Lower Arm Score | Wrist Score | Posture Score B | Coupling Score | Score B

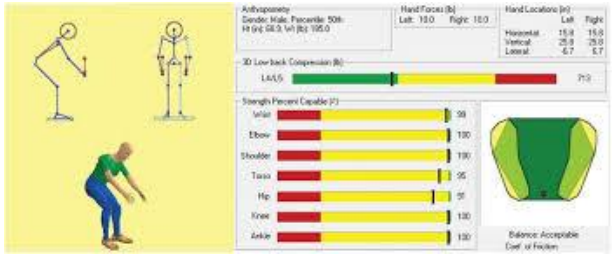
Original Worksheet Developed by Dr. Alan Hedge. Based on Technical note: Rapid Entire Body Assessment (REBA), Hignett, McAtamney, Applied Ergonomics 31 (2000) 201-205

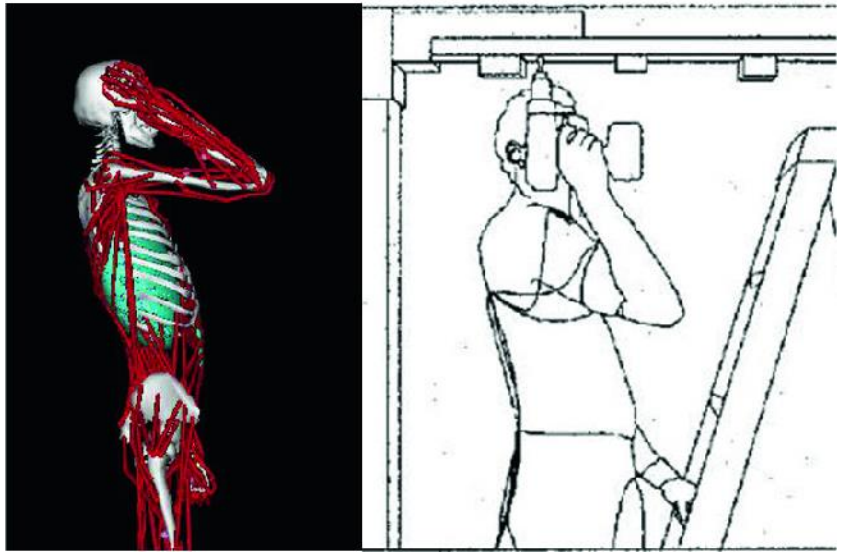
👎 Empirical, subjective
👎 Joint level

EMG-based approach



👍 Direct and accurate
👎 Only feasible in lab





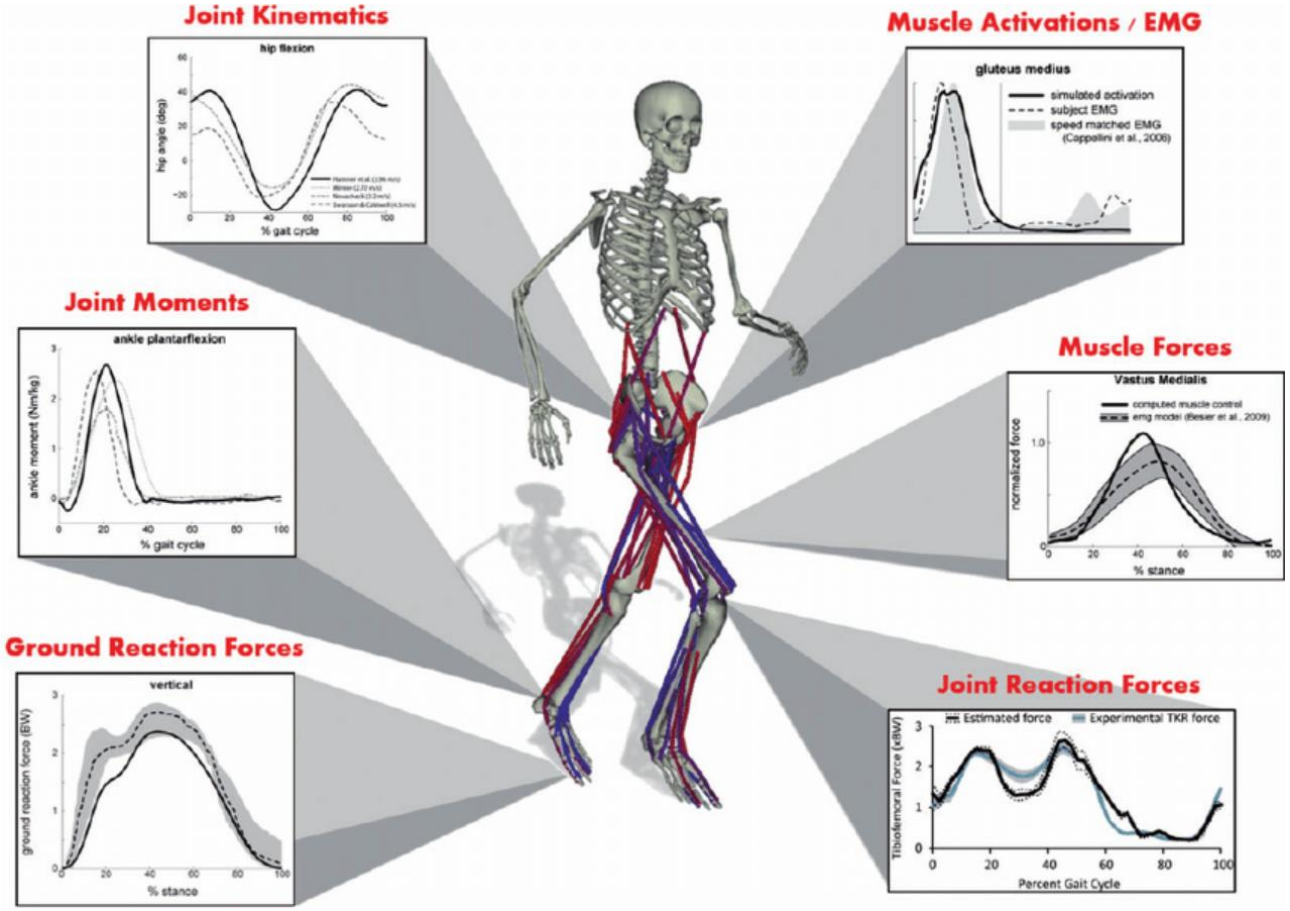
👍 Muscle level
👎 Complex to conduct

Introduction to Biomechanics

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Biomechanics is the study of how mechanical forces are applied to and generated by human, and the effects of these forces on human



What does biomechanics study:

- Internal/external forces act on body
- How biological tissues respond to loading
- How different body parts interact with each other

Application of biomechanics:

- Sports performance
- Injury prevention
- Rehabilitation
- Biomedical devices
- Ergonomics

Overview

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Session 1: Introduction to OpenSim

Objectives:

✓ Learn basics of OpenSim

Hands-on Activities:

- Run basic simulations on OpenSim using examples

Session 2: Introduction to OpenSim API

Objectives:

✓ Learn basics of OpenSim Python API

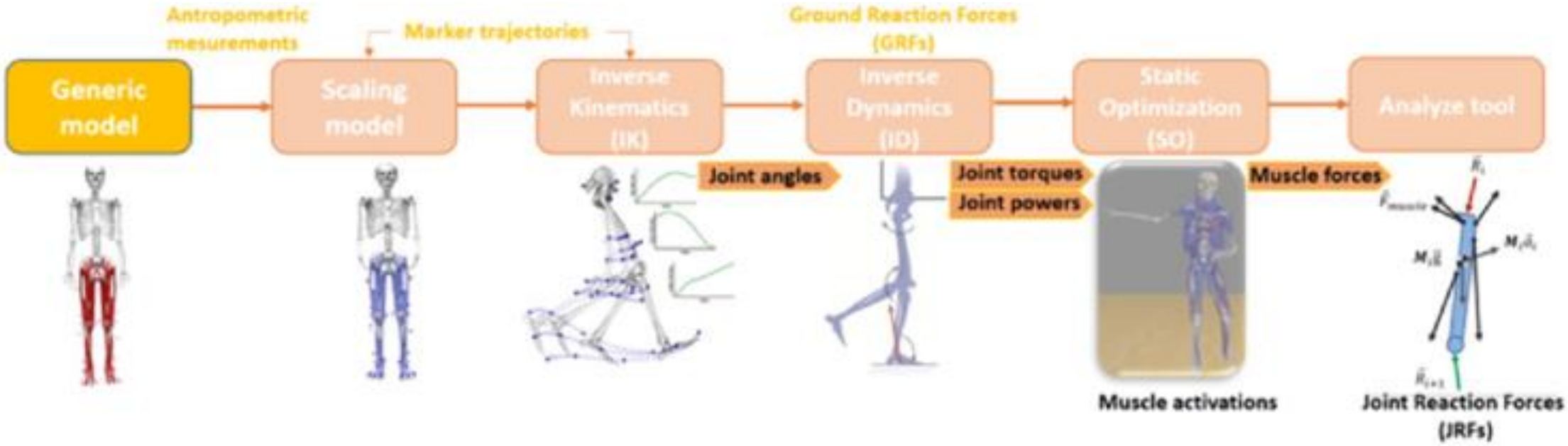
Hands-on Activities:

- Run basic simulations with OpenSim API

Session 3: Simulate human-exoskeleton

- Construct human-exoskeleton through OpenSim Python API
- Run simulations on human-exoskeleton system

Workflow of OpenSim simulation



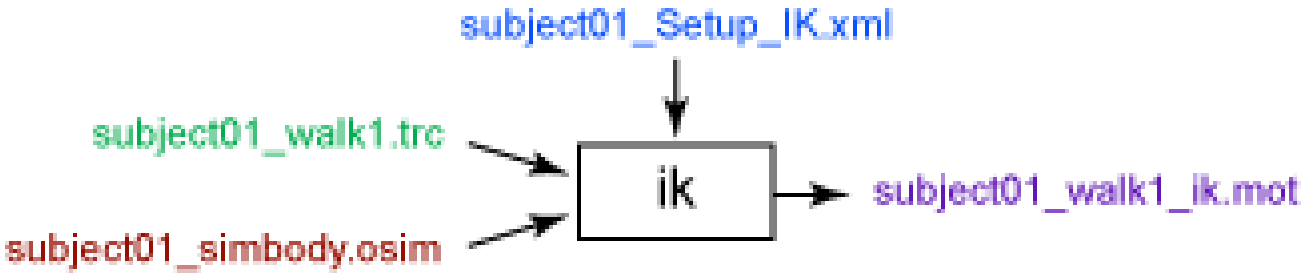
For more info, visit: [User's Guide - OpenSim Documentation - OpenSim]

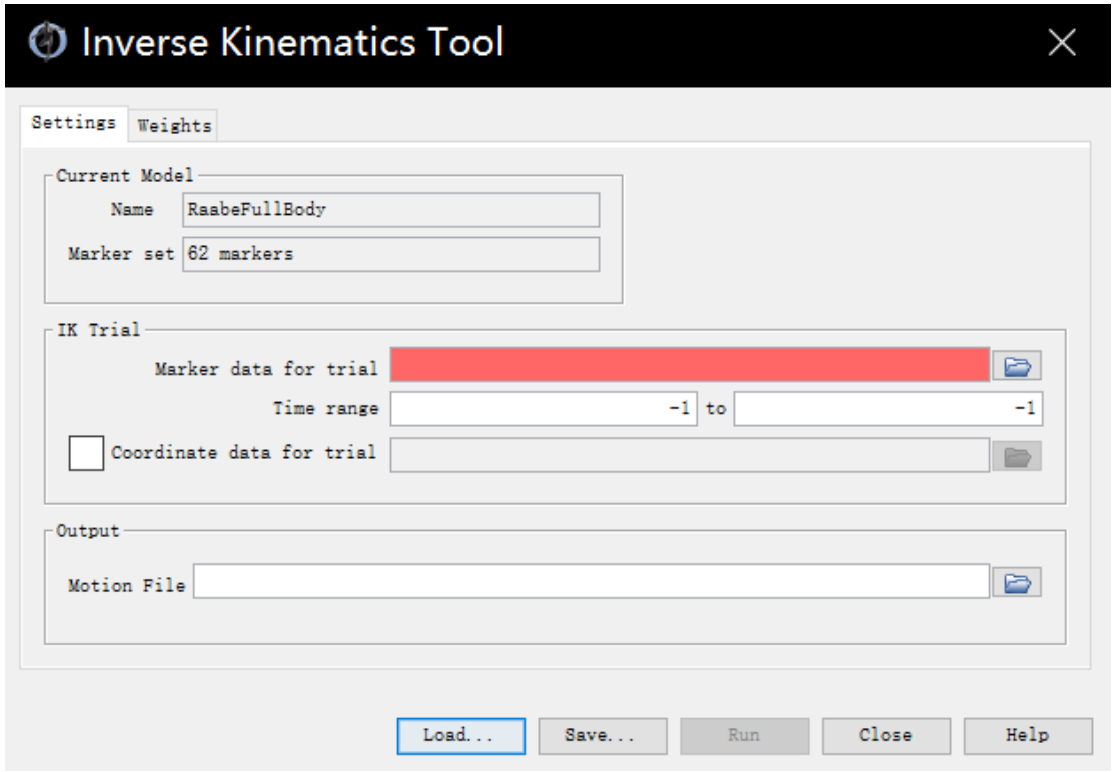
Inverse kinematics in OpenSim

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IK inputs and outputs





How inverse kinematics work

$$\min_{\mathbf{q}} \left[\sum_{i \in \text{markers}} w_i \left\| \mathbf{x}_i^{\text{exp}} - \mathbf{x}_i(\mathbf{q}) \right\|^2 + \sum_{j \in \text{unprescribed coords}} \omega_j \left(q_j^{\text{exp}} - q_j \right)^2 \right]$$

$$q_j = q_j^{\text{exp}} \text{ for all prescribed coordinates } j$$

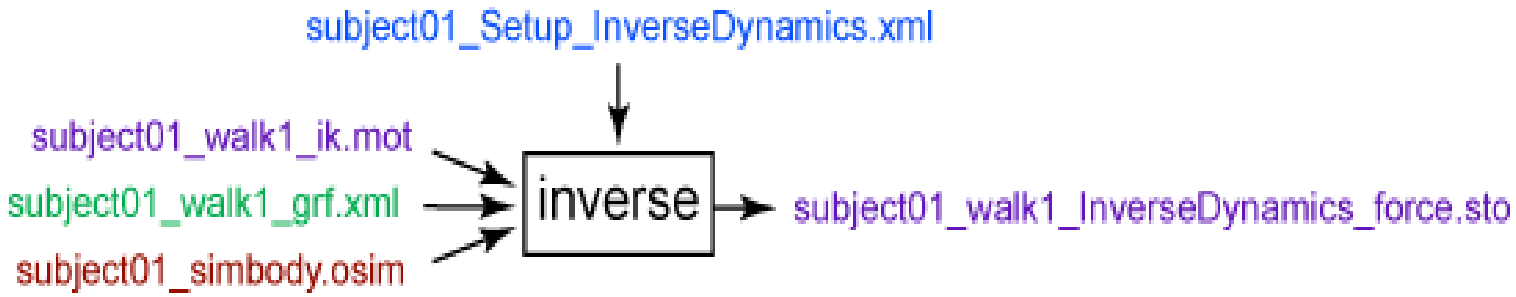
Obtain a kinematic trajectory (joint angles) that minimizes the distance between the model-defined markers and marker trajectories obtained from motion capture experiment

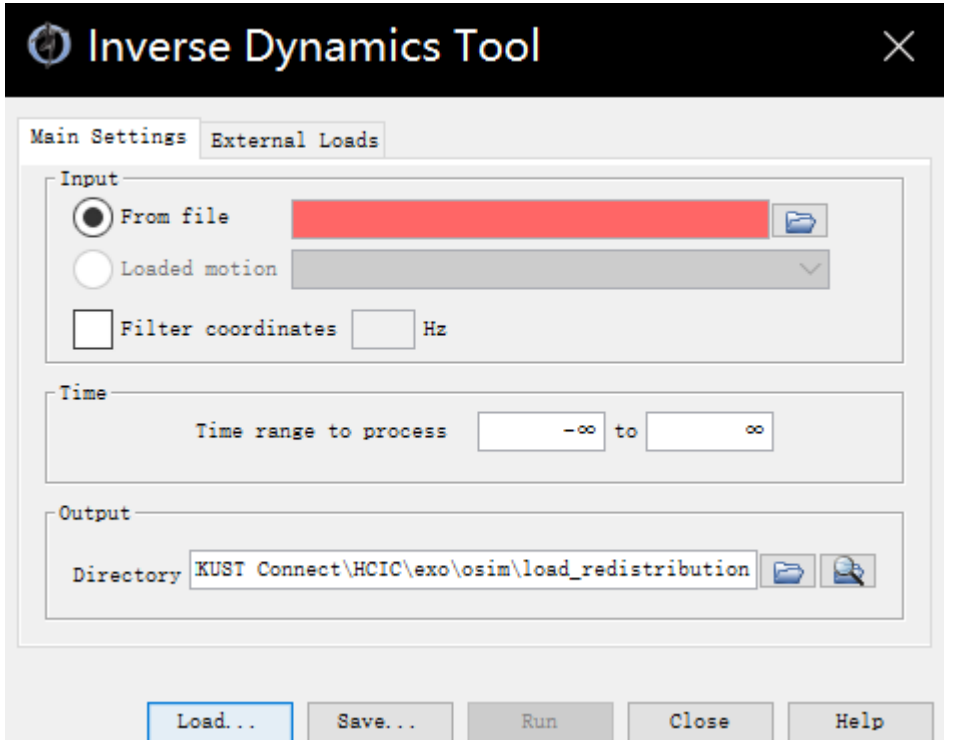
Inverse kinematics in OpenSim

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ID inputs and outputs





How inverse dynamics work

$$\underbrace{\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}}+\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})+\mathbf{G}(\mathbf{q})}_{\text{knowns}} = \underbrace{\tau}_{\text{unknowns}}$$

$q, \dot{q}, \ddot{q} \in R^N$ Positional parameters

$M(q) \in R^{N \times N}$ System mass matrix

$C(q, \dot{q}) \in R^N$ Centrifugal force

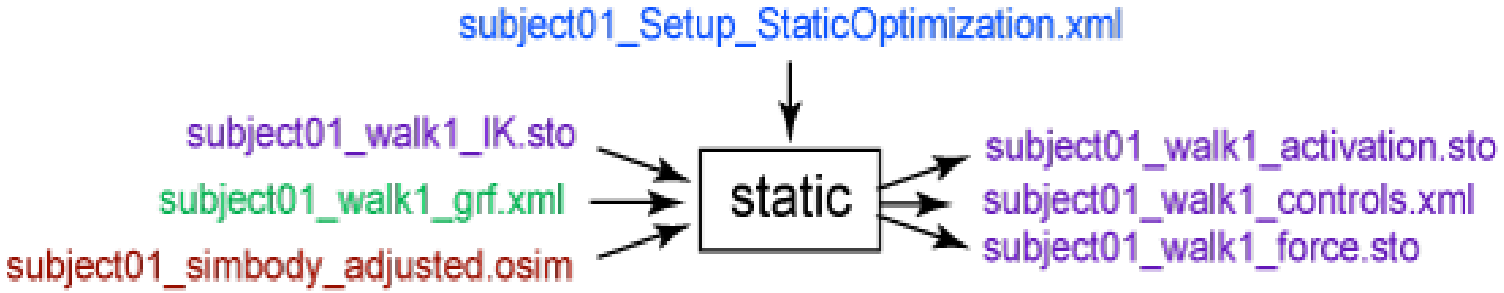
$G(q) \in R^N$ Gravitational force

$\tau \in R^N$ Generalized forces

Yield the net joint moments/torques and forces required to create the motion

Static Optimization in OpenSim

SO inputs and Outputs



At each time step, solve:

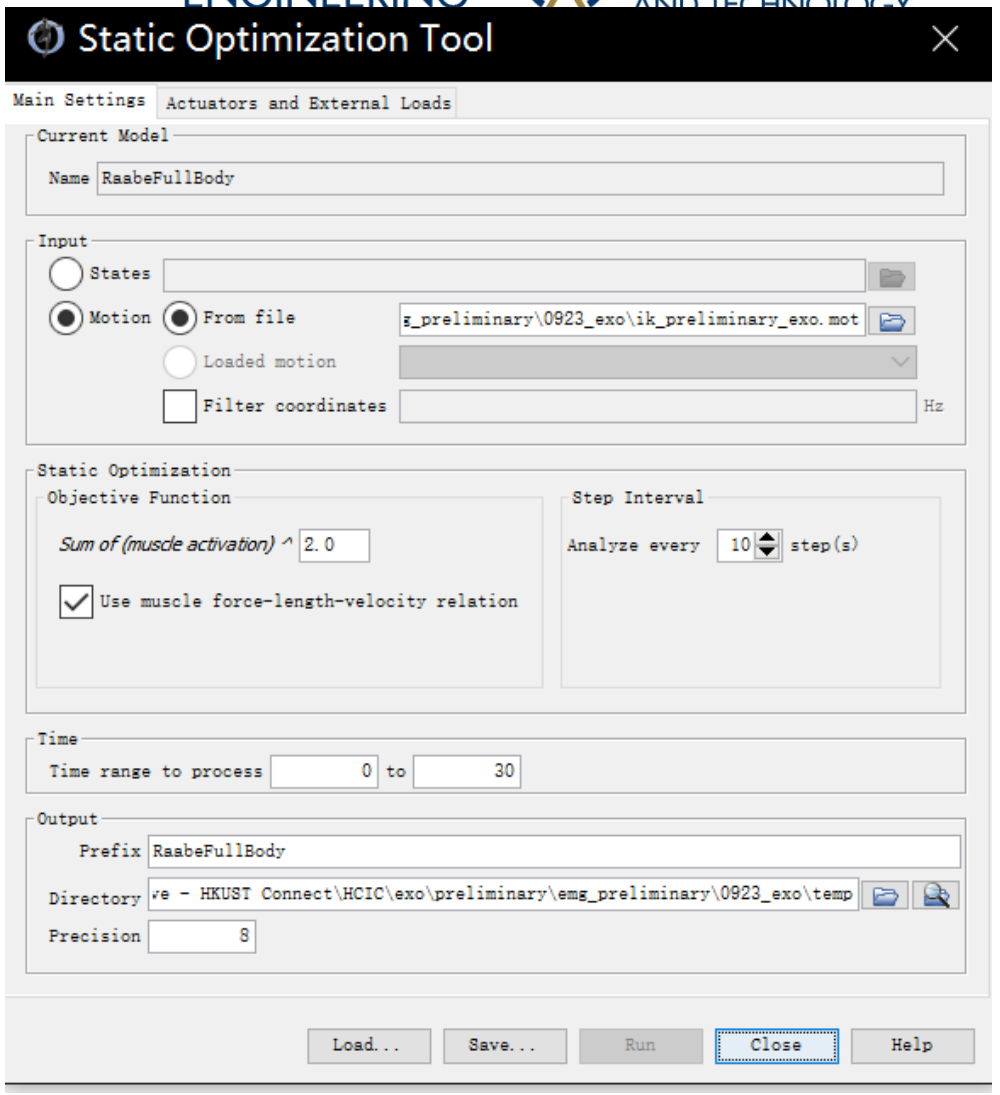
$$\min \sum_{i=1}^{n} a_i^p$$

where a_i is the muscle activation

Subject to joint moment constraint

$$\sum_{i=1}^{n} r_i F_i = \tau_{\mathrm{ID}}$$

r_i is the muscle moment arm, F_i is muscle force, τ_{ID} is joint moment from ID



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Introduction

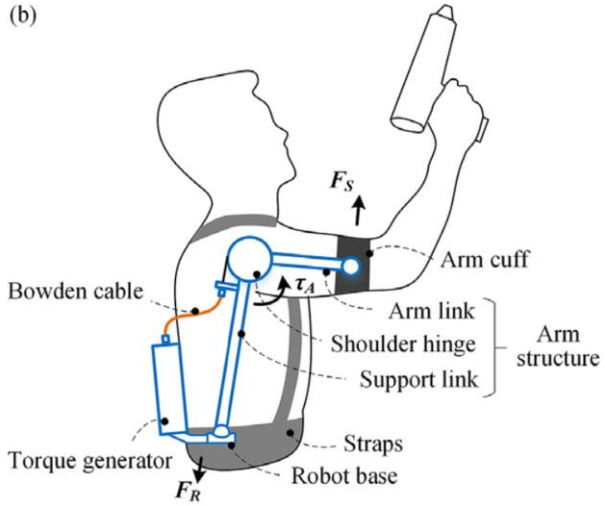
WMSD mitigation - exoskeletons

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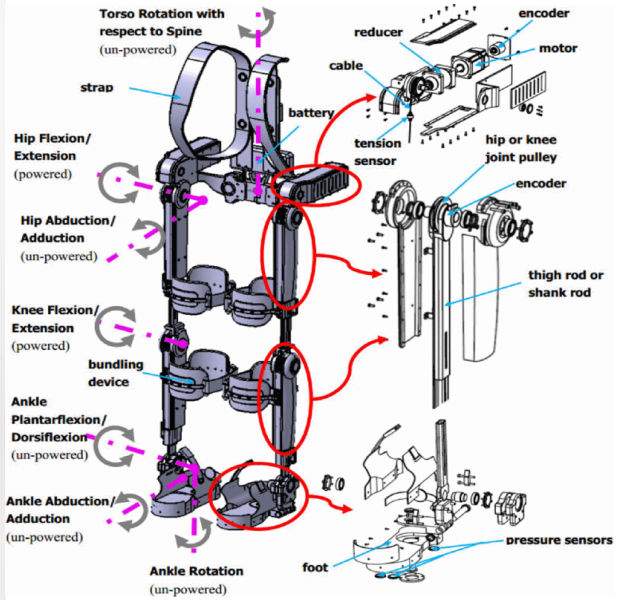
Passive exoskeletons





Active exoskeletons

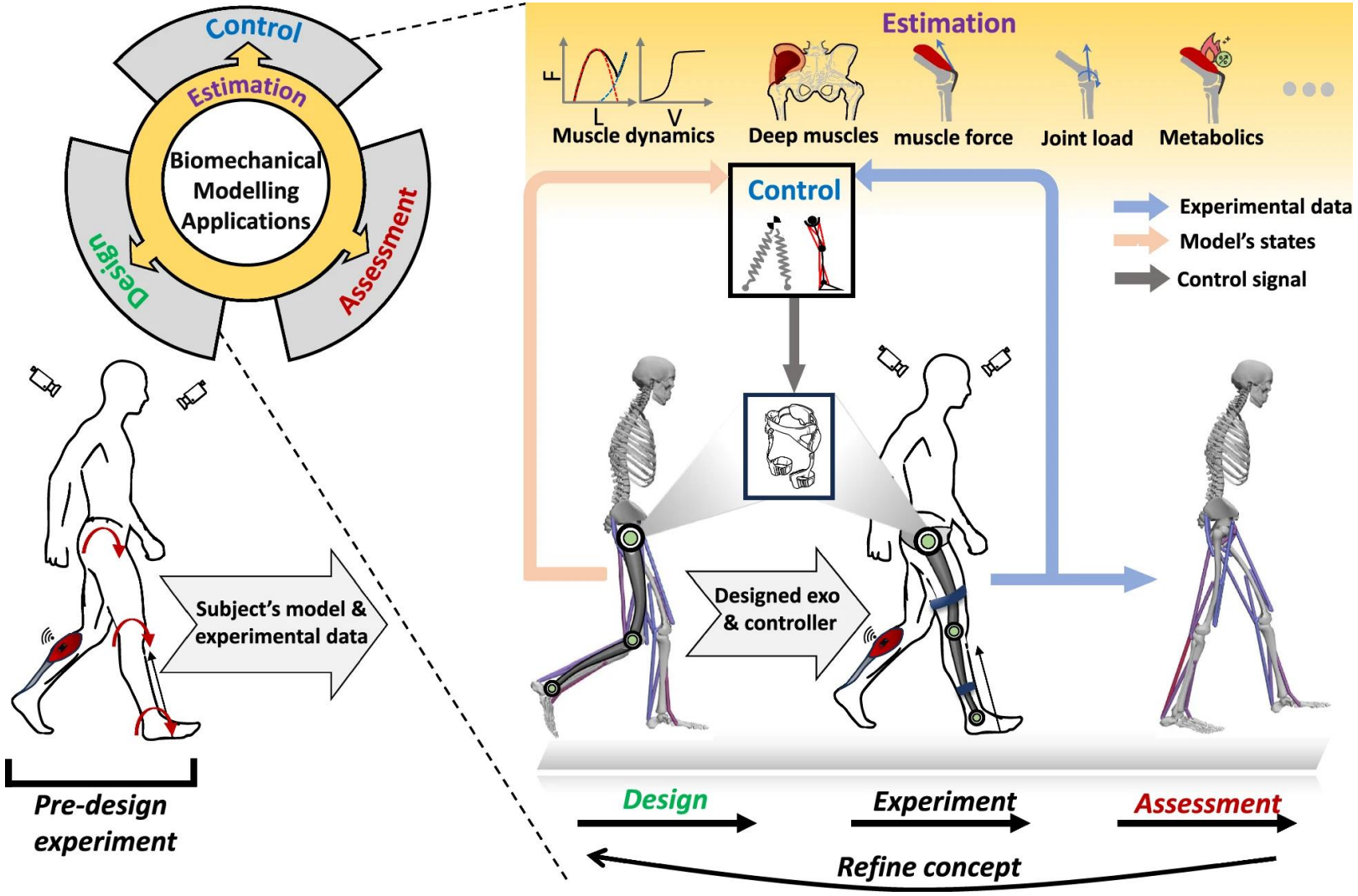




Exoskeletons can reduce physical burden of construction tasks

But how to evaluate the impact of exoskeletons to human biomechanical system?

Human-exoskeleton simulation in OpenSim



Use of biomechanical simulation in exoskeleton applications

- Design
- Control
- Assessment
- Estimate

Firouzi et al, 2025

Biomechanical simulation for exoskeleton assessment

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Exoskeleton TRY ON

Build a simple OpenSim-exoskeleton model through OpenSim Python API

Assignment

Plot the **moment** on ‘lumbar_bending’ and **muscle force** on ‘ercspn_r’ with spring **stiffness** 4000 N/m